

Static Tucker: benchmark tables

H100 80GB HBM3, BF16, Llama 257M, seq 1024

Capacity: 16,384 tokens/step, columns: microbatch $\times$ accumulation

1 Forward, mc

Метод	1 $\times$ 16	2 $\times$ 8	4 $\times$ 4	8 $\times$ 2	16 $\times$ 1
Dense AdamW	133.7	69.5	38.4	37.7	35.6
Dense Muon	136.6	68.8	38.8	37.7	36.1
Static Tucker (5 streams)	448.0	222.4	116.4	75.8	73.3

2 Backward, mc

Метод	1 $\times$ 16	2 $\times$ 8	4 $\times$ 4	8 $\times$ 2	16 $\times$ 1
Dense AdamW	227.0	108.7	79.7	75.7	71.2
Dense Muon	231.4	108.8	79.9	75.6	71.7
Static Tucker (5 streams)	868.4	430.5	213.2	134.0	124.8

3 Optimizer, mc

Метод	1 $\times$ 16	2 $\times$ 8	4 $\times$ 4	8 $\times$ 2	16 $\times$ 1
Dense AdamW	1.8	1.8	1.8	1.8	1.8
Dense Muon	42.8	40.6	33.5	34.3	34.7
Static Tucker (5 streams)	586.7	582.3	585.3	574.8	574.7
Tucker sequential (old)	444.4	424.4	398.1	446.8	439.4

4 Полный шаг, мс

Метод	1×16	2×8	4×4	8×2	16×1
Dense AdamW	364.2	181.5	120.9	116.0	109.5
Dense Muon	413.0	219.8	153.0	148.4	143.4
Static Tucker (5 streams)	1909.1	1237.4	919.0	786.6	775.3

5 Peak CUDA allocated memory, ГБ

Метод	1×16	2×8	4×4	8×2	16×1
Dense AdamW	3.27	4.41	6.61	11.02	19.28
Dense Muon	2.96	4.10	6.28	10.69	18.98
Static Tucker (5 streams)	4.23	6.49	10.73	19.42	36.26

6 Production: microbatch 32 × accumulation 4

Метод	Forward	Backward	Optimizer	Шаг	Peak, ГБ
Dense AdamW	274.303	547.362	1.750	824.186	37.441
Dense Muon	276.340	551.348	35.009	863.574	37.136
Static Tucker (5 streams)	557.133	942.166	571.316	2073.215	71.539

7 Runs

Метод	ID
Commit	69e569c
Capacity	1m-mpi-job-0583f75d-0fa8-4f7b-8e5e-18d8fae2f90a
Production	1m-mpi-job-f405573b-c021-46cc-bad7-5103d709b023